

Portfolio Optimization

A Machine Learning Approach using XGBoost

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Executive Summary

- Project goal: Predict the 10 best performing stocks for the following month with the intention of beating the index OMXS30.
- Macrodata, stock prices and engineered ML features used for analysis.
- XGBoost model used for analysis with Ridge Regression as a baseline.
- The model was checked for data leakage and look ahead bias, which was not found.

KEY RESULTS

20.24% return over the benchmark

Sharpe ratio of **0.67**

Model return: **51.37%**

OMXS30 return: **31.13%**

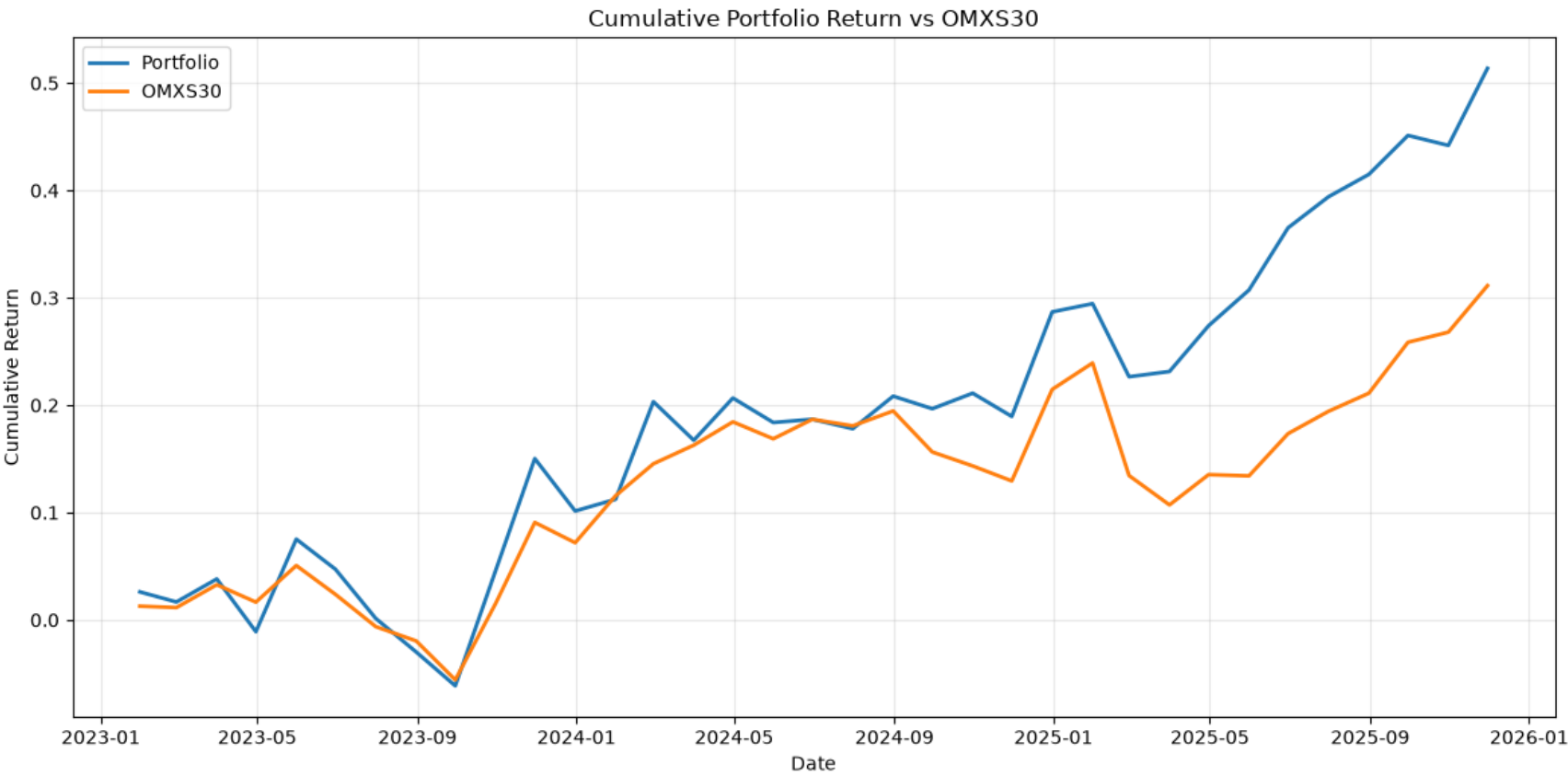
Performance and Portfolio Metrics

PORTFOLIO RETURN

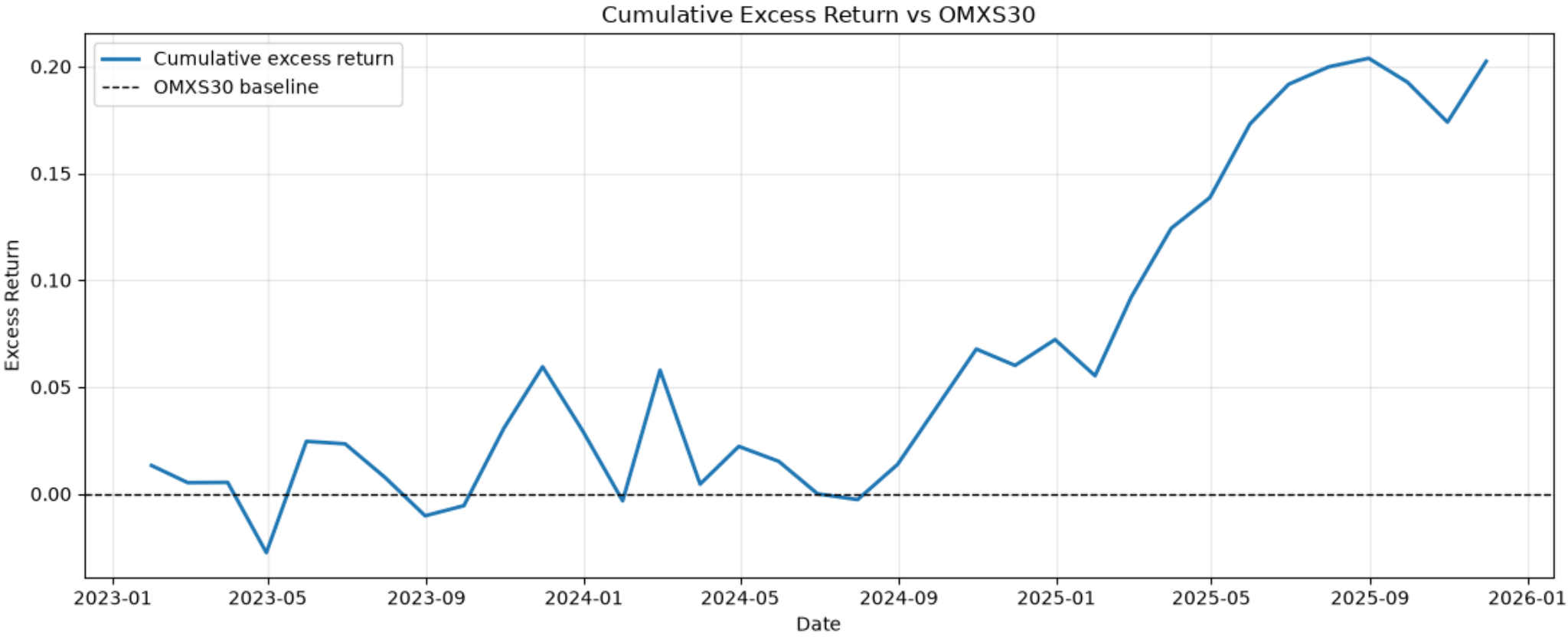
51.37%

OMXS30 RETURN

31.13%



Performance and Portfolio Metrics



Validation Measures

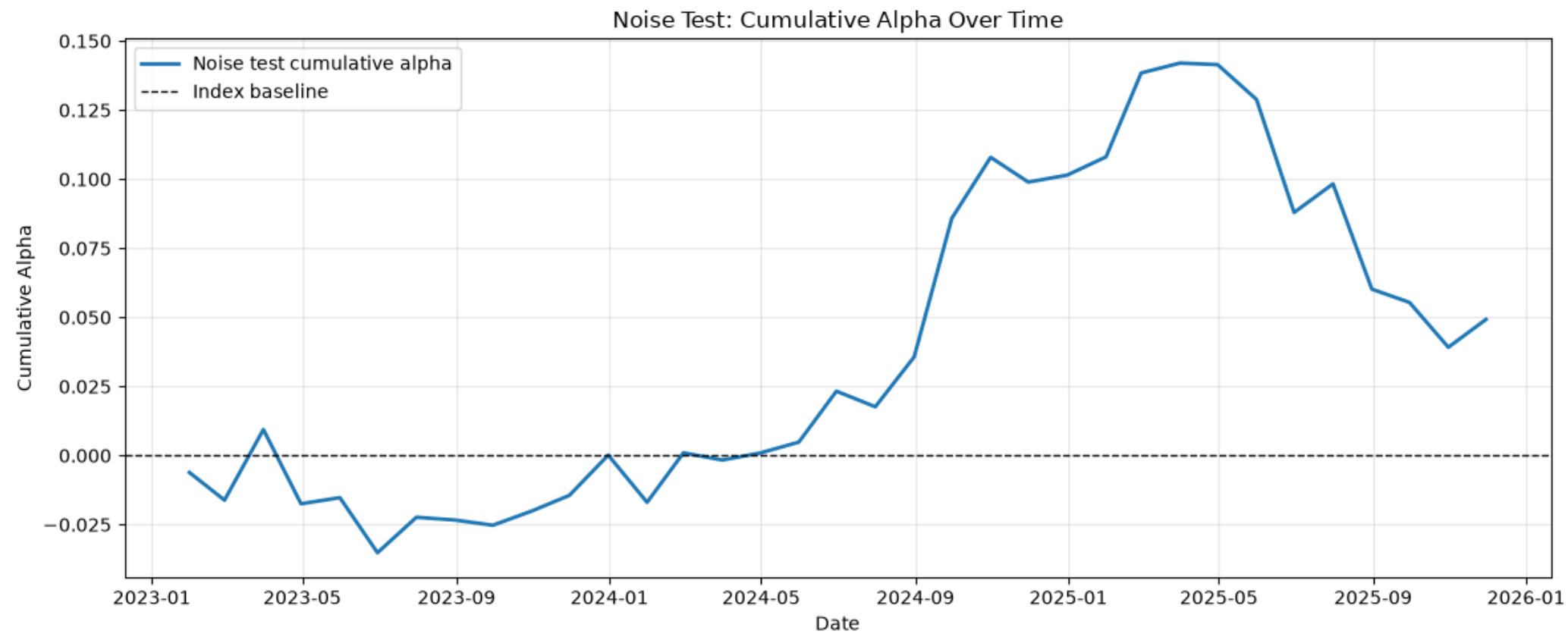
Look-Ahead Verification

Compared the model's expected returns for individual stocks against their actual future performance. The clear difference between them confirms that the model does not have access to data in the future.

The Noise Test

Tested the model using completely randomized data. Because the model's Alpha decreased to 4.92%, it proves the original returns come from real market signals, not a broken or leaking backtest.

Noise Test



Limitations and Future Adjustments

Portfolio Allocation

Move from equal 10% weight to optimization methods like Mean-Variance or Risk Parity to better manage asset risk and correlation.

Transaction Costs & Slippage

Factor in trading fees and market spread.

Alternative Models

Evaluate other AI models such as LightGBM, LambdaMART ranking or LSTM networks.

Dynamic Rebalancing

Transition from a strict monthly schedule to risk-triggered rebalancing.